greater detail below.

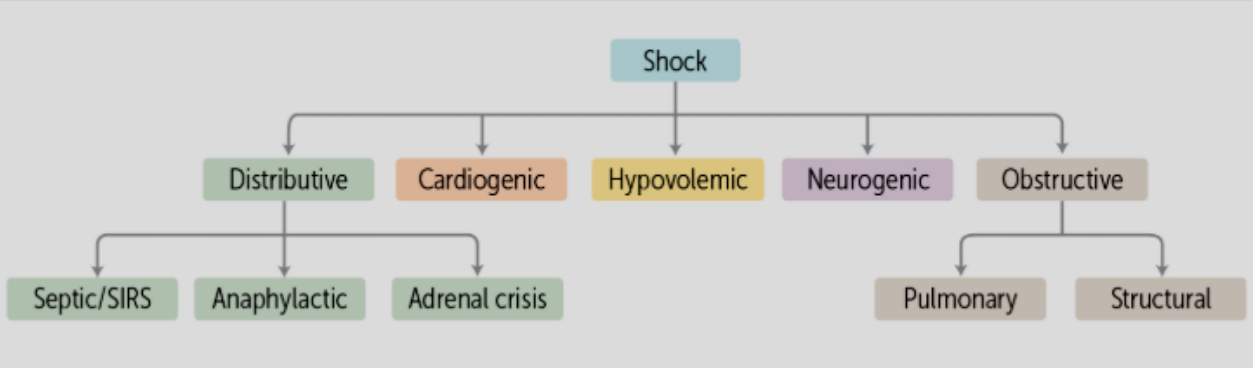

Figure 1 Types of shock

Clinical Presentation of Shock (LO2)?

The clinical presentation of shock combines common features across all types with distinctive characteristics that help identify the underlying mechanism. Early recognition of these patterns is crucial for directing appropriate diagnostic and therapeutic interventions.

Regardless of etiology, patients in shock typically exhibit several hallmark findings:

- **Hypotension:** Systolic blood pressure below 90 mmHg or mean arterial pressure below 65 mmHg (may be absent in early compensated shock). This is associated with syncope or presyncope, especially orthostatic.
- **Tachycardia** is nearly universal (except in certain conditions like neurogenic shock or in patients on beta-blockers) as the heart attempts to maintain cardiac olume.
- **Altered mental status** reflects cerebral hypoperfusion and is a sensitive indicator of shock severity. It can range from anxiety and restlessness in early stages to confusion, lethargy, and eventually coma as cerebral perfusion deteriorates.
- Shortness of breath, with increased **respiratory rate** to compensate for metabolic acidosis.
- **Urine output decreases** as the kidneys experience reduced perfusion, with oliguria (output <0.5 mL/kg/hr) being a sensitive marker of shock.
- **Skin findings** often include cool, blue, or painful fingers and toes, though specific patterns vary by shock type.

Mnemonic

Use this definition of shock to remember the five types: **N**ew **C**ells **D**own and **O**ut (from) **H**ypoxia = **N**eurogenic, **C**ardiogenic, **D**istributive, **O**bstructive, and **H**ypovolemic.

What is shock?

Table 1 Summarizes the clinical findings seen in shock.

Table 1 Clinical findings seen in shock

System or location	Signs
Nervous	Mental status change (eg, confusion, loss of consciousness)
Cardiovascular	Sweating, ↑ heart rate, ↓ blood pressure
Respiratory	↑ Respiratory rate
Extremities	Cyanosis
Renal	↓ Urine output (oliguria)

What are some signs and symptoms of severe shock?

What Physiologic Parameters Do We Follow in a Patient with Shock (LO3)?

The assessment and management of shock rely on measuring key physiologic parameters that reflect cardiovascular function and tissue perfusion. These measurements guide diagnosis, treatment decisions, and evaluation of therapeutic responses.

Mean arterial pressure (MAP) represents the average pressure driving blood flow through the systemic circulation during the cardiac cycle. These pressures are routinely measured by using a standard blood pressure cuff. It is calculated as below where DBP and SBP are diastolic and systolic blood pressure. Note pulse pressure can also be used and is (SBP-DBP).

$$MAP = DBP + \frac{1}{3}(SBP - DBP)$$

A MAP of at least 65 mmHg is generally considered necessary for adequate perfusion of vital organs, though higher targets may be appropriate for patients with chronic hypertension.

Cardiac output (CO) is the volume of blood pumped by the heart per minute, normally 4-8 L/min. Cardiac output can be measured non-invasively through echocardiography or invasively via thermodilution with a pulmonary artery catheter.

cardiac output = HR × SV

Stroke volume is decreased by:

- Decreased **preload**, which is a decrease in the pressure of the heart at the end of diastole and can be estimated by the end-diastolic volume (EDV)
- Increased **afterload**, which is an increase in the pressure resisting blood flow out of the heart and is usually determined by the SVR (arteriolar resistance) but also estimated by the mean arterial pressure
- Decreased **myocardial contractility**, which is a decrease in the strength of the heart muscle contractions and is independent of preload and afterload

Systemic vascular resistance (SVR) or total peripheral resistance (TPR) is the total resistance to blood flow and is mainly determined by arteriolar constriction, which is nervous system:

SVR = MAP/cardiac output

Arterial Oxygen Content

Because shock is ultimately caused by failed oxygenation of the blood or failed extraction of oxygen by the tissue, we often calculate the amount of oxygen delivered to the tissues. Oxygen delivery is determined by the product of cardiac output (CO) and arterial oxygen content (CaO_2): $DO_2 = CO \times CaO_2$.

Arterial oxygen content is primarily determined by hemoglobin concentration (Hb) and its O_2 carrying capacity of 1.36, oxygen saturation (SaO_2), and to a lesser extent, dissolved oxygen (PaO_2) and its solubility of 0.03. This relationship is expressed as below:

$$CaO_2 = \mathrm{Hb} \times 1.36 \times (SaO_2)_a + (PaO_2 \times 0.03)$$

The latter term of dissolved oxygen in the blood is very small, so the content equation can be simplified as:

$$CaO_2 = \mathrm{Hb} \times 1.39 \times (SaO_2)_a$$

C_aO_2 and $(SaO_2)_a$ are measured in millimeters of mercury (mm Hg). The $(SaO_2)_a$ is measured using routine arterial blood gas analysis, and the Hb is measured as part of the complete blood count.

Arterial oxygen content could fall because of:

- Decreased cardiac output
- Anemia (decreased Hb)
- Decreased oxygen saturation of arterial blood (hypoxemia), usually due to lung disease

Sometimes, the oxygen content is measured in both arterial and venous blood, and the difference is calculated (termed the **arteriovenous oxygen difference**). This value is sometimes useful diagnostically. For example, in septic shock, the tissues often do not extract much oxygen, making for a small difference between arterial and venous oxygen content (small arteriovenous oxygen difference).

What Are the Five Types of Shock (LO4)?

Shock can be classified into five primary types based on the underlying pathophysiologic mechanism.

Each type presents with a characteristic pattern of alterations in cardiac output (CO), systemic vascu (EV) that guides both diagnosis and management strategies (Figure 3). Each of these types of shock are described in detail in the sections below.

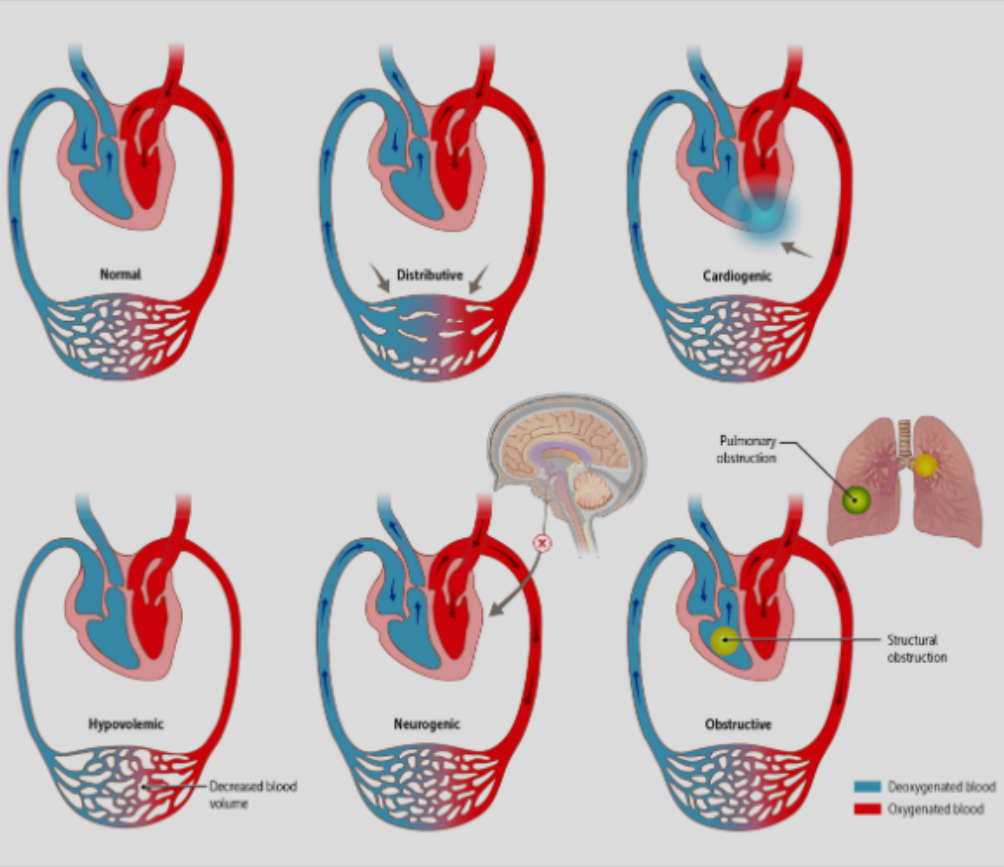

Figure 3 Normal state and trigger insults of the five types of shock

Distributive Shock: Decreased Systemic Vascular Resistance (blood pooling)

Distributive shock results from pathologic vasodilation and increased vascular permeability, leading to maldistribution of blood flow and relative hypovolemia. In distributive shock, extensive systemic vasodilation causes a decrease in SVR and

The primary culprit in distributive shock is **low SVR** due to vasodilation. The low SVR forces the body to compensate and increase the cardiac output in an attempt to maintain the MAP. The body increases cardiac output by increasing both heart rate (tachycardia) and stroke volume, each driven by the sympathetic nervous system. Blood is pumped rapidly out of the heart, leading to a decreased EDV (preload)

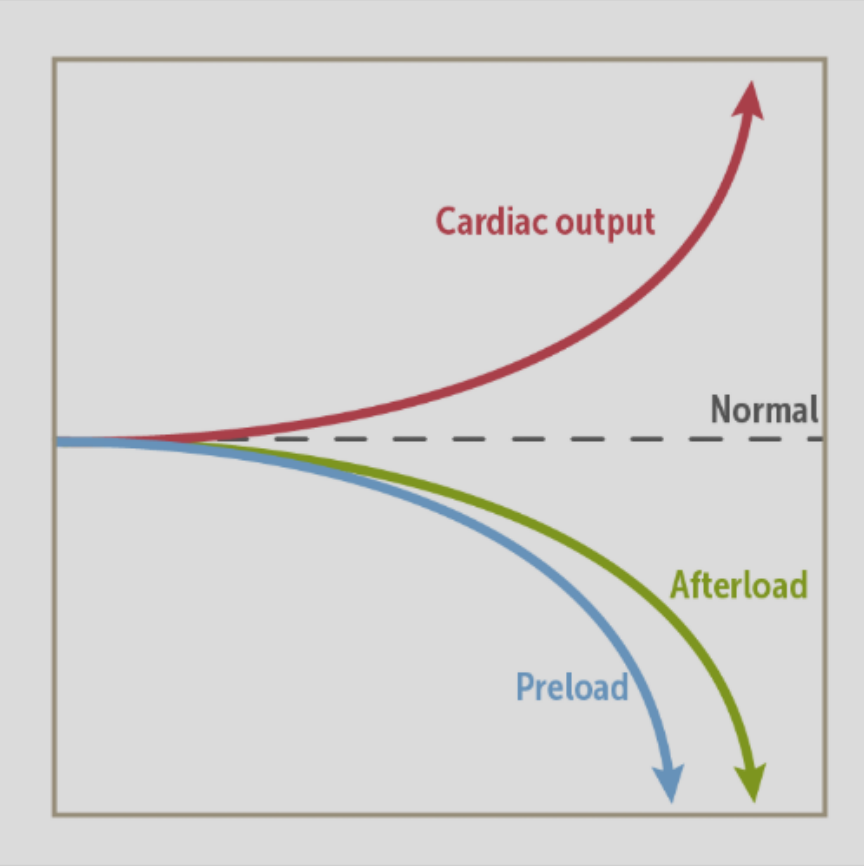

Figure 4 Distributive shock

What causes this vas ses:

Clinically, the patient will be hypotensive, but the peripheral vasodilation may cause the skin to be warm even at the extremities, which are usually cold in the other types of shock.

What happens to MAP during distributive shock and why?

Cardiogenic Shock: Pump failure

In cardiogenic shock, blood pressure drops because of the heart's failure to pump blood through the circulatory system (**decreased cardiac output**). Some of the causes can include myocardial ischemia and infarction, cardiomyopathies, valvular disease, and arrhythmias.

To maintain blood pressure in the face of low cardiac output, the body must raise the SVR as a compensatory response. This arteriolar constriction leads to high afterload in car spite the low blood pressure, which can be clinically confusing! Just remember that blood pressure support blood pressure but ultimately increase myocardial oxygen demand and afterload, exacerbating cardiac dysfunction.

Clinically, the patient has cool extremities, tachycardia, weak pulses, elevated JVD, and an S3 heart sound. The EDV (preload) rises due to the inability of the heart to adequately pump the blood, leading to pulmonary edema. Pulmonary edema results in dyspnea, crackles and hypoxemia (from pulmonary edema),

> **CLINICAL CORRELATION**
>
> To treat cardiogenic shock, one possibility is an inotropic drug to increase cardiac contractility, although these generally also increase the risk of dangerous arrhythmias. Vasodilators decrease systemic vascular resistance and, in theory, should decrease the amount of work demanded by the failing heart. However, the blood supply to the heart itself requires adequate systemic vascular resistance to sustain the diastolic blood pressure, since the coronary arteries only fill during diastole. Therefore, vasodilators should not be used to treat cardiogenic shock in isolation; some drugs like dobutamine and milrinone act as both inotropes and vasodilators but still run the risk of causing decompensation of the blood pressure in a patient already in shock.

What are some of the causes of cardiogenic shock?

the severity of shock until decompensation occurs. However, because of low preload, the cardiac output is generally low (Figure 6).

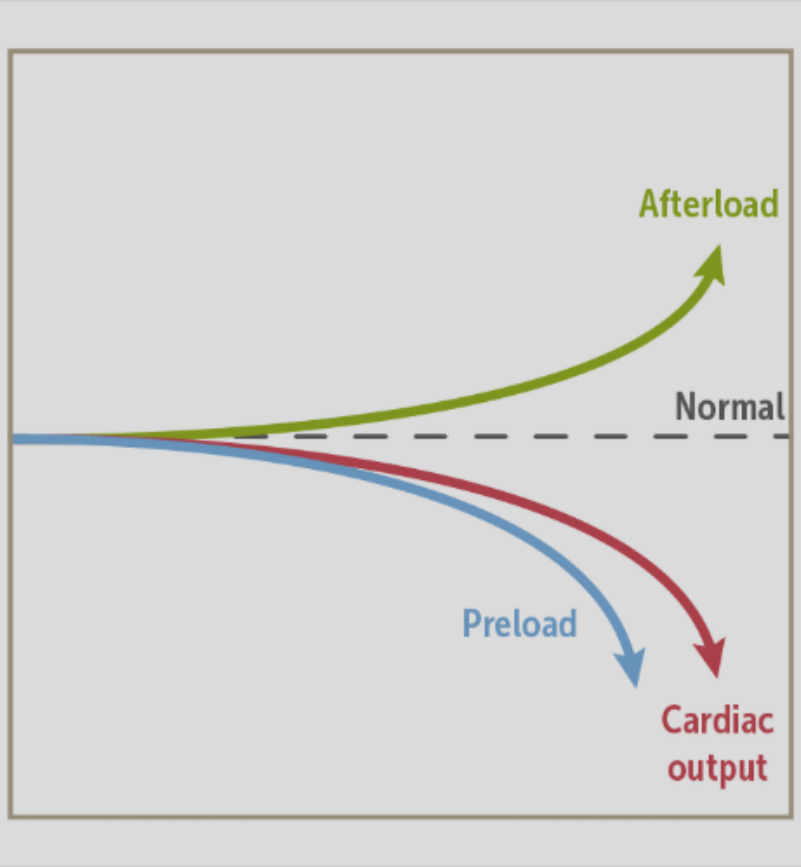

Figure 6 Hypovolemic shock

Clinically, the patient has cold extremities (from vasoconstriction) and may appear pale due to the hemorrhage. A source of bleeding (trauma, GI) is often obvious. The severity of hemorrhagic shock correlates with the volume of blood lost, with clinical manifestations typically appearing after 15-30% of blood volume is depleted

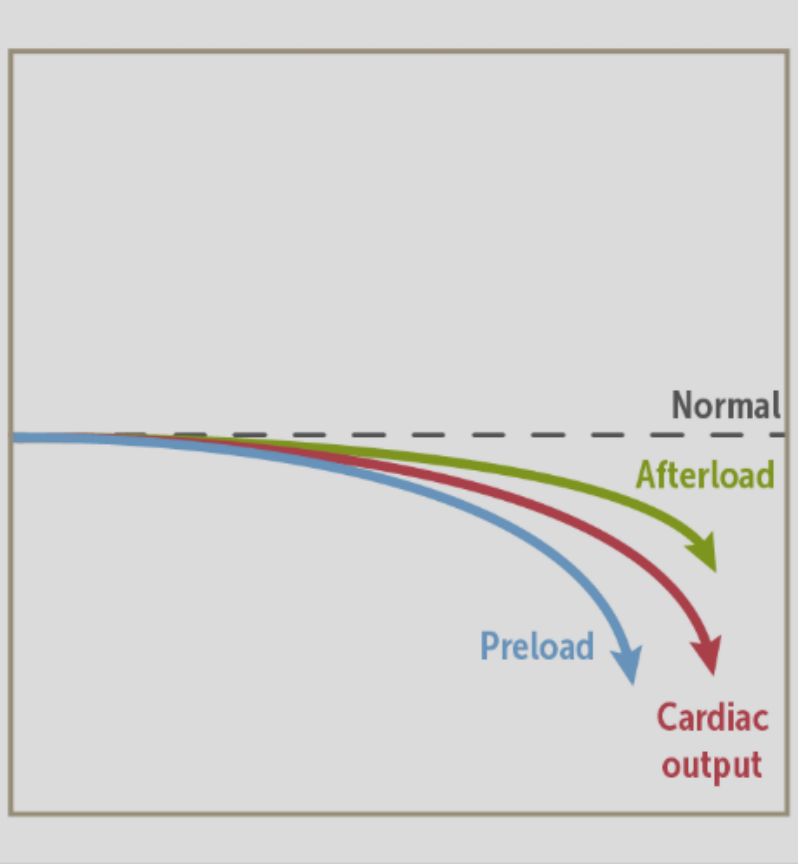

Figure 7 Neurogenic shock

Clinically, neurologic signs and symptoms and a history of trauma to the brain or spinal cord are often seen. The classic triad includes hypotension, bradycardia (instead of the expected tachycardia), and warm, dry skin below the level of injury due to loss of sympathetic tone. It's also the only type of shock in which the patient does not have tachycardia since the sympathetic nervous system cannot mount that response. Some authors place this under the umbrella of distributive shock since the SVR drops, but the cardiac response is quite diff

What is the definition of constrictive pericarditis?

How Do Patients With Constrictive Pericarditis Present? (LO #1)

Patients with constrictive pericarditis usually present with signs and symptoms relating to the impairment of the heart's diastolic filling; the impairment is caused by the heart's encasement by the inelastic pericardium. With impaired filling, cardiac output decreases, and patients may report dyspnea or fatigue on exertion. In addition, there might be signs of fluid overload, such as peripheral edema, ascites, and pleural effusions. This fluid overload occurs because blood returning to the heart cannot be accommodated by the stiff pericardium, leading to a backup of fluid into the systemic and pulmonary circulation.

What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Constrictive Pericarditis

Last updated March 2, 2026 · 12 min read · ScholarRx

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Editors: Jordan Finn, Russell Johnson

Adapted from: Constrictive Pericarditis

Listen to this Brick
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Learning Objectives (4) · Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of constrictive pericarditis.
2. Explain the pathophysiology of constrictive pericarditis.
3. Interpret clinical findings to properly diagnose constrictive pericarditis.
4. Outline the rditis.